

*MA.7.DP.1.3***Benchmark****MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.**

Example: O'Neill's Pillow Store made 600 pillows yesterday and found that 6 were defective. If they plan to make 4,300 pillows this week, predict approximately how many pillows will be defective.

Example: A school district polled 400 people to determine if it was a good idea to not have school on Friday. 30% of people responded that it was not a good idea to have school on Friday. Predict the approximate percentage of people who think it would be a good idea to have school on Friday from a population of 6,228 people.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.AR.3.1, MA.7.AR.3.2
- MA.7.AR.4.5

Terms from the K-12 Glossary

- Categorical Data
- Population (in Data Analysis)
- Proportional Relationships
- Random Sampling

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.4
- MA.6.DP.1.3, MA.6.DP.1.4

Next Benchmarks

- MA.8.DP.1.1
- MA.912.DP.1.4

Purpose and Instructional Strategies

In grade 6, students described data using measures of center and variation. In grade 7, students use samples to compare measures of center and variation in data sets, as well as use samples to make a generalization about the population from which the sample was taken. In grade 8, students will use bivariate data to study proportional and linear relationships and make predictions with lines of fit. In high school, students will estimate a population total, mean or percentage using data from a sample survey and develop a margin of error through the use of simulation.

- Instruction includes helping students understand that since there is always variability in collecting samples. These generalizations, or predictions, can only be estimates of what we expect to see from the greater population.
- Use real-world scenarios to explain that random sampling is needed when we need to find information about a population that is too large or too difficult to measure completely (*MTR.7.1*).
 - As in the second example for this benchmark, the school district polled 400 people because it would be difficult, and perhaps costly, to poll all 6,228 people efficiently. Therefore, we can analyze a sample that is representative of the population to get an idea of what may be happening with the larger group.

- Students have done more precise work with proportional relationships in MA.7.AR.4, but in this benchmark, students will use the same proportional reasoning to make predictions or find estimates of what may be happening in a population that is too large or cumbersome for us to measure completely.
- Instruction includes having students make predictions about what the population measures will be, based on the sample (*MTR.6.1*).
- Instruction uses manipulatives or simulations to have students collect their own set of data (*MTR.2.1*).
- As in the first example for this benchmark about defective pillows, students can pull a random sample from a bag of chips that have been strategically marked D for defective or have no marking for no defects. They can use their proportion of defective chips to predict how many might be in the entire bag. Comparisons can be made across the different groups in the room to see how close the estimates were to the actual values (*MTR.4.1*).

Common Misconceptions or Errors

- Students may mistake part to total as part to part, which would give an incorrect ratio when setting up their proportion. To address this misconception, use percentages or counts out of 100 to help illustrate this more clearly.
- Students may not understand what random sampling is or why it is important. To address this misconception, allow students to collect random samples and make comparisons across groups to show they are not exact, but representative, of the larger population.

Strategies to Support Tiered Instruction

- Teacher provides instruction focused on color-coding and labeling the different categories based on the sample and the population when setting up a proportional relationship to ensure corresponding parts are placed in the corresponding positions within the proportion.
 - For example, based on a random sample of 150 people, 23 people stated that they preferred to go grocery shopping on Saturday morning. If one wants to make a prediction on how many people, out of a town of 4500 people, who prefer to go shopping on Saturday morning, the proportion below can be used.

$$\frac{23 \text{ Saturday shoppers}}{150 \text{ shoppers}} = \frac{x \text{ Saturday shoppers}}{4500 \text{ shoppers}}$$

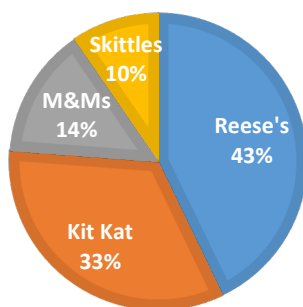
Students can also make the connection to multiplicative relationships between shoppers and Saturday shoppers as shown below.

$$\frac{23 \text{ Saturday shoppers}}{150 \text{ shoppers}} = \frac{x \text{ Saturday shoppers}}{4500 \text{ shoppers}}$$

$\xrightarrow{\times 30}$
 $\xleftarrow{\times 30}$

- Teacher uses percentages to help illustrate the difference between part to part and part to total more clearly. Use percentages or counts out of 100 to help illustrate this more clearly.
- Instruction includes providing students with an example of random sampling and biased sampling in a context that is relevant to students.
 - For example, Branden at Sunshine Middle School wants to predict if pizza should be served at the Fall Festival and Johnny suggests sampling 20 students in his 7th grade class. Amanda suggests it would be better to sample 20 random students from all grade levels at Sunshine Middle School. Since Branden wants to ensure that the sampling is not biased, he chooses Amanda's plan since the prediction is for the whole school and not just for one class at the school.
- Instruction includes the use of problems with percentages or counts of 100.
 - For example, Mr. Smith surveyed students at his school last year to determine which candy bar they preferred. His results are shown in the pie chart below. Students can discuss how they can use this information to make a prediction about how many students prefer a certain candy bar this school year.

■ Reese's ■ Kit Kat ■ M&Ms ■ Skittles



- Teacher allows students to collect random samples and make comparisons across groups to show they are not exact, but representative of the larger population.

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.5.1)

A random sample of the 1,200 students at Moorsville Middle School was asked which type of movie they prefer. The results are compiled in the table below:

Action	Comedy	Historical	Horror	Mystery	Science Fiction
15	12	3	10	4	6

- Part A. Use the data to estimate the total number of students at Moorsville Middle school who prefer horror movies.
- Part B. Use the data to estimate the total number of students at Moorsville Middle school who prefer either mystery or science fiction movies.
- Part C. Suppose another random sample of students was drawn. Would you expect the results to be the same? Explain why or why not.

Instructional Task 2 (MTR.4.1, MTR.7.1)

A constitutional amendment is on the ballot in Florida, and it needs **at least 60%** of the vote to pass. The editor of a local newspaper wants to publish a prediction of whether or not the amendment will pass. She hires ten pollsters to each ask 100 randomly selected voters if they will vote yes.

Poll number	Yes votes
1	51
2	68
3	58
4	43
5	65

Poll number	Yes votes
6	56
7	45
8	59
9	71
10	63

Should the newspaper's prediction be that the amendment *will pass* or that it *will not pass*? Explain your reasoning.

Instructional Items

Instructional Item 1

A research group is trying to determine how many alligators are in a particular area. They tagged 30 alligators and released them. Later, they counted 12 alligators who were tagged out of the 150 they saw. What can the research group estimate is the total population of alligators in that area?

Instructional Item 2

The local grocery store is going to donate milk and cookies to an upcoming middle school event. They surveyed 150 students in the school to determine which type of milk they preferred and recorded the results below.

Cow's Milk	Soy Milk	Almond Milk
82	25	43

If there are 950 students in the school, how much soy milk should the store plan to donate?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive*